

Touseef Ur Rehman

Machine Learning Engineer | Full-Stack Web Developer

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Professional Summary

Final-year Computer Science student (GPA 3.27 / 4.0) with two completed internships in Machine Learning and Full-Stack Web Development. Skilled in PyTorch, TensorFlow, Hugging Face Transformers, Flask, FastAPI, and React—building AI/ML pipelines, NLP classifiers, and production web platforms. Seeking a Software Engineering or ML role where I can contribute immediately.

Education

BS Computer Science

HITEC University Taxila — Taxila, Punjab, Pakistan

Sep 2022 – Sep 2026

GPA: 3.27 / 4.0

Coursework: Machine Learning, Deep Learning, Data Mining, NLP, Computer Vision, Software Engineering, Database Systems, Algorithms & Data Structures.

Technical Skills

Languages	Python, C++, JavaScript (ES6+), HTML5, CSS3, React
ML / AI	Scikit-learn, TensorFlow, Keras, PyTorch, Hugging Face Transformers, NLTK, Pandas, NumPy, Matplotlib, MLflow
NLP / Transformers	BERT, DistilBERT, RoBERTa, XLNet, GPT-2, fine-tuning, tokenizers, ensemble classifiers
Computer Vision	YOLOv8, OpenCV, object detection, real-time inference
Backend & APIs	Flask, FastAPI, Node.js, Express, REST APIs, BeautifulSoup, Selenium
Databases	MySQL, SQLite, Supabase (PostgreSQL), schema design, query optimisation
Tools	Git, GitHub, Jupyter, Google Colab, VS Code, Render, Railway, XAMPP

Work Experience

Machine Learning Intern

Elevo — Remote (Egypt)

Sep 2025

1 Month

- Delivered a full computer vision pipeline using PyTorch and OpenCV—covering data preprocessing, model design, training, evaluation, and inference—under a strict one-month deadline.
- Built predictive models for automated loan approval, applying feature engineering and cross-validation to reduce financial risk decision errors.
- Developed K-Means clustering pipelines for customer segmentation, transforming raw transactional data into actionable behavioural cohorts for business intelligence.
- Optimised a multi-class forest cover classification model through hyperparameter tuning; built a MovieLens collaborative filtering recommendation engine using matrix factorisation.

Website Developer Intern

RedFort360 — Wah Cantt, Pakistan

May 2025 – Aug 2025

4 Months

- Engineered fully responsive React UIs with hooks and component-driven architecture for multiple live client projects, delivering smooth cross-browser experiences.
- Built Flask REST APIs integrated with Supabase PostgreSQL, following scalable backend patterns aligned with production standards.
- Deployed applications on Render and Railway with proper environment configuration, build pipelines, and uptime documentation.
- Worked in agile sprints alongside senior engineers, meeting all client milestones while maintaining professional Git and code review practices.

Key Projects

AI-Powered Marine Pollution Detection SystemYOLOv8 · LSTM · FastAPI · React — Final Year Project

- Built a real-time marine plastic debris detection platform using YOLOv8 trained on custom underwater imagery, delivering high-confidence bounding-box predictions on live video streams.
- Implemented LSTM time-series models to forecast ocean pollution concentration from environmental sensor data, enabling proactive monitoring and early-warning outputs.
- Delivered a full-stack React / FastAPI web platform with live detection feeds and an interactive pollution heatmap dashboard, managing the complete ML lifecycle independently.

Transformer-Based Cyberbullying DetectionBERT · RoBERTa · XLNet · GPT-2 · Python — CS-405

- Fine-tuned BERT, DistilBERT, RoBERTa, XLNet, and GPT-2 on a labelled social media dataset under CPU-only constraints, reproducing results from a peer-reviewed research paper.
- Designed an ensemble soft-voting classifier combining all five models, achieving accuracy that exceeded every individual model baseline.
- Delivered a complete Jupyter Notebook pipeline and a formal report with training curves, evaluation metrics, and cross-model comparison tables.

AI-Powered Public Opinion AnalyzerNaive Bayes · NLTK · TF-IDF · Flask

- Developed a Flask web application with a Naive Bayes classifier for real-time multi-class sentiment analysis on AI-related content, with a full NLP preprocessing pipeline (tokenisation, stemming, TF-IDF) and probability-based confidence scoring.

Smart Faculty Attendance Management SystemReact · Vite · Supabase · PostgreSQL

- Built a real-time teacher attendance portal with role-based access control, live database synchronisation, date-filtered reporting, and exportable admin records—deployed as a live production application for Govt High School Khanpur, KPK.

HITEC University Library Management SystemFlask · MySQL · Bootstrap 5 · REST API

- Delivered a production-ready library system with JWT authentication, book inventory CRUD, borrowing history, overdue tracking, and a full admin dashboard on a normalised MySQL schema with a responsive Bootstrap 5 frontend.

Additional Information

Languages	English (Professional), Urdu (Native)
Availability	Immediately available—full-time, hybrid, or fully remote
Open To	On-site (Pakistan), Remote (Global), Relocation for postgraduate studies (China, USA)
Interests	ML systems design, NLP research, computer vision, AI for social good, scalable web engineering
Repositories	All projects publicly available at github.com/touseef7878

Certifications

Computer Vision Specialization	University of Colorado Boulder
Google AI Essentials Specialization	Google
Prompt Engineering Specialization	Vanderbilt University
MLOps Specialization	Duke University
Full Stack Web Development	Amazon
Back-End Apps with Node.js & Express	IBM